

Software Requirements Specifications

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Prepared by



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Introduction

Traditional time-tracking tools act as passive digital notebooks, leaving developers, managers and admins to manually reconcile hours logged against project budgets and deadlines. Without a shared, real-time view of who is working on what, teams will frequently discover that a project is over budget or behind schedule only after the fact, once a timesheet has already been approved and the week is closed. Momently bridges the gap between raw time entries and actionable project intelligence; it gives Developers a faster way to log and submit their hours and gives Managers and Admins a live, role-scoped view of project progress, team hours and budget consumption, without waiting on a manual report. The system is role-aware from the ground up, the same screen shows a Developer their own assigned work, a Manager their whole workspace and an Admin the entire organisation, without separate applications.

User Stories / User Characteristics

User Management & Authentication

- As a user, I want to enter my company email and password and hit submit so that the system authenticates me and grants access to my role-appropriate dashboard.
- As a user, I want to fill out the registration form with my name, company email and password so that the system creates my new account and I am assigned the default Developer role.
- As a user, I want to verify my email address after registration so that I can unlock full access to the system.
- As a user, I want to log in with my Google or Microsoft account so that I don't need a separate password for Momently.
- As a user with MFA enabled, I want to enter a 6 digit code from my authenticator app after my password so that my account has a second layer of protection.
- As a user, I want to reset my password using a link sent to my email so that I can regain access if I forget it.

- As a user, I want to select the log-out option so that the system securely ends my active session and returns me to the login screen.

Workspace & Role Management

Guest

- As a guest, I want to see a waiting screen and have an admin notified so that I know I haven't been forgotten while I wait to be added.

Manager

- As a manager, I want to view my workspace so that I can see all projects and members in my team.
- As a manager, I want to invite a user to my workspace so that I can add new team members.
- As a manager, I want to transfer workspace ownership to another user so that I can hand over responsibility when needed.
- As a manager, I want to assign Project Manager roles to any workspace member so that they can have project permissions for a specific project

Administrator

- As an admin, I want to view all workspaces in the system so that I have full visibility over the entire application.
- As an admin, I want to delete a workspace so that I can remove it from the system when it is no longer needed.
- As an admin, I want to invite a user to any workspace so that I can add members to any team or organisation.
- As an admin, I want to change the owner of a workspace so that I can transfer ownership to another user.

Timesheets

Developer

- As a developer, I want to view my current week's time entries grouped by task so that I can see my daily and weekly totals at a glance.
- As a developer, I want to filter my timesheets by status (Draft, Submitted, Approved, Rejected) so that I can quickly find the ones I still need to act on.
- As a developer, I want to submit my week's draft entries as a single timesheet so that my manager can review and approve them.
- As a developer, I want to see my manager's rejection reason inline on a rejected timesheet so that I know exactly what to correct before resubmitting.

Manager

- As a manager, I want to view all submitted timesheets in my workspace so that I can view them efficiently
- As a manager, I want to approve a submitted timesheet so that it becomes locked in as finalised
- As a manager, I want to reject a submitted timesheet with a written reason so that the developer knows exactly what to fix.
- As a manager, I want an AI-generated evidence summary alongside a submitted timesheet so that I can spot entries with no supporting GitHub, Jira or calendar activity before I approve.

Administrator

- As an admin, I want to view all timesheets across all workspaces so that I can monitor the overall submission and approval status of the organisation.
- As an admin, I want to approve any timesheet in any workspace so that I can finalise timesheets when needed.
- As an admin, I want to reject any timesheet with a reason so that I can provide feedback on submissions.

Task Management

Developer

- As a developer, I want to view all tasks assigned to me so that I know what work I still need to complete.
- As a developer, I want to filter and search my tasks by status, priority, or title so that I can focus on what matters right now.
- As a developer, I want to update a task's status so that my current progress is reflected on screen.

Manager

- As a manager, I want to view all tasks within a project so that I can see what work is in progress.
- As a manager, I want to create tasks and assign them to team members so that work is properly distributed.
- As a manager, I want to set task priorities and due dates so that the team knows what to focus on first.
- As a task creator, I want to set task priority and due date so that the team knows what to focus on first.

Project Management

Developer

- As a developer, I want to view the projects I am personally assigned to so that I know which projects I can log time against.
- As a developer, I want to see my own hours logged and each project's progress so that I understand how my work contributes to the whole.

Manager

- As a manager, I want to view every project in my workspace, not just the ones I'm personally on, so that I have full visibility over my team's workload.
- As a manager, I want to see each project's team roster and each member's hours logged so that I can spot over- or under-loaded team members.
- As a manager, I want to see a project's progress against its budgeted hours so that I can tell if a project is on track.
- As a manager, I want to see each project's budgeted cost and hourly rate so that I can manage cost, not just time.

Administrator

- As an admin, I want to view every project across every workspace in the organisation, not just my own, so that I have a single, org-wide view of all work in flight.
- As an admin, I want to see the same team roster, hours, progress and cost detail as a manager, but for any project in the organisation, so that I can support any team without needing to be added to it.

Leave Request

Developer

- As a developer, I want to request for leave so that my timesheets, time entries don't affect my insights against being flagged as a non-worker.
- As a developer, I want to upload supporting documents so my manager can approve my request.
- As a developer, I want to update or cancel my own pending leave request so that I can fix a mistake or change my mind before it's reviewed.

AI Insights & GitHub Integration

Developer

- As a developer, I want to see my weekly summary, productivity score, task-switching rate, delivery forecast and any flagged anomalies so that I understand my own working pattern.

- As a developer, I want to switch between an overall view and a per-project view of my insights so that I can see whether a specific project is where my time or risk is concentrated.

Manager / Admin

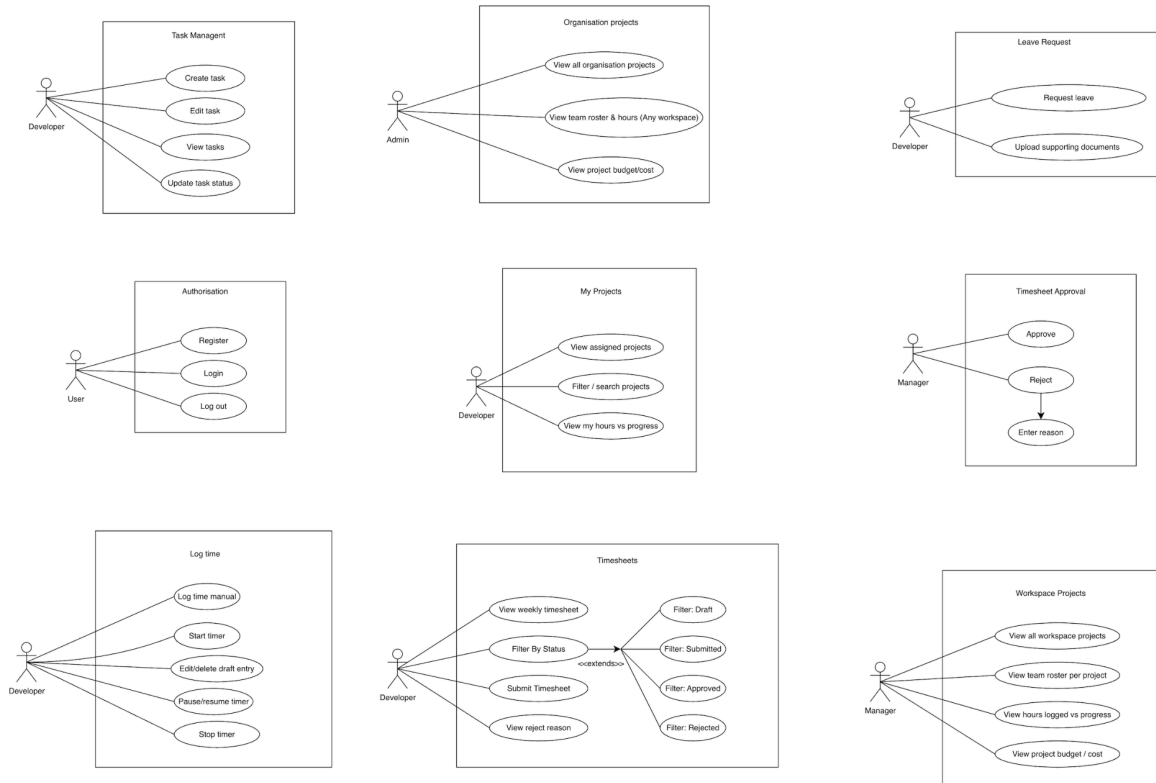
- As a manager or admin, I want to see workspace-wide burnout risk flags, per-project delivery forecasts and team productivity trends so that I can act before a problem becomes a deadline miss.
- As a manager or admin, I want a team weekly summary written in plain English so that I don't have to read every developer's raw numbers myself.
- As a user, I want to mark an insight as resolved so that it stops showing up as an open flag once I've dealt with it.

Integrations

any user

- As a user, I want to connect my GitHub account so that my commit activity can be compared against my logged hours.
- As a connected user, I want to manually sync my GitHub activity so that the comparison reflects my most recent commits.
- As a user, I want to connect Jira so that I can see my ticket breakdown and logged versus estimated hours without leaving Momently.
- As a user, I want to connect Google Calendar so that my tracked time can be compared against my scheduled meetings.

High-level Use Case Diagram

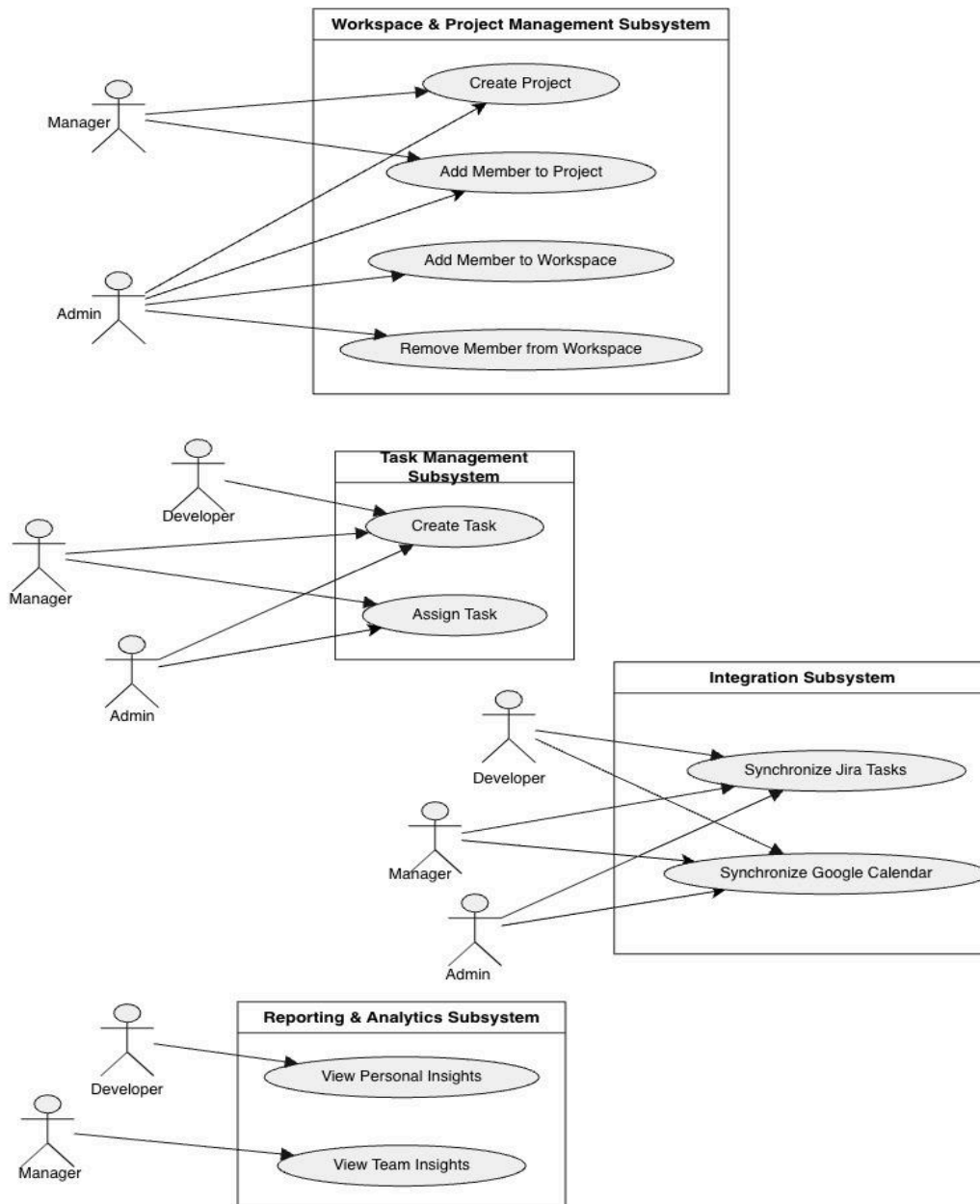


Use Cases

Each use case below states **TUCBW** (This Use Case Begins When) and **TUCEW** (This Use Case Ends When) as recommended by the lecture slides and textbook.

Note: A Manager inherits developer as a Manager can be a developer in a different project

Authorisation



Register

- TUCBW the user, not yet authenticated, submits the registration form with first name, last name, company email and password.
- TUCEW the system creates the new account (default role: Developer) and the user is authenticated or routed to log in with their new credentials.

Login

- TUCBW the user enters their company email and password and hits submit.

- TUCEW the system authenticates the credentials, issues a JWT and grants access to the user's role-appropriate, guarded routes.

Complete MFA challenge

- TUCBW a login attempt succeeds for a user with MFA enabled.
- TUCEW the user enters a valid 6 digit authenticator code and receives their JWT, or is refused after an invalid or expired code.

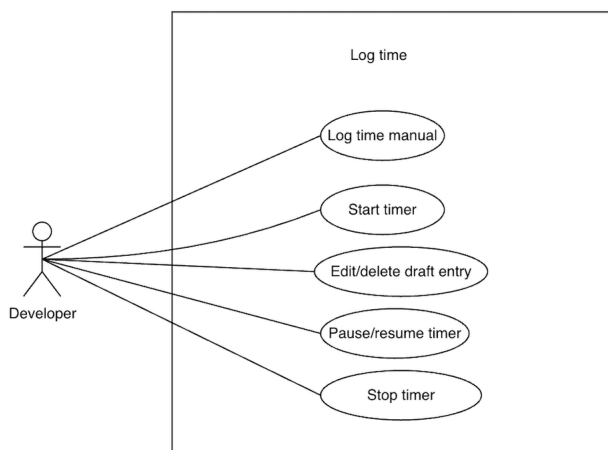
Log in with Google or Microsoft

- TUCBW the user chooses Google or Microsoft on the login screen.
- TUCEW the provider token is validated, the account is created or linked and a JWT is issued.

Log out

- TUCBW the user selects the log-out option from the app menu.
- TUCEW the system clears the active session/token and returns the user to the login screen.

Log Time (Developer)



Log time (manual)

- TUCBW the user opens the Log Time screen and opens the manual-entry panel.

- TUCEW the new time entry is saved as a draft and appears in the current day's list.

Start timer

- TUCBW the user selects a project and task and clicks "Start Timer."
- TUCEW an active timer session begins and elapsed time displays and updates every second.

Pause / resume timer

- TUCBW the user pauses an active timer.
- TUCEW elapsed time is held exactly and resuming continues counting from that point without loss.

Stop timer

- TUCBW the user clicks "Stop" on an active timer.
- TUCEW a draft time entry is created automatically from the elapsed duration, with no further user action required.

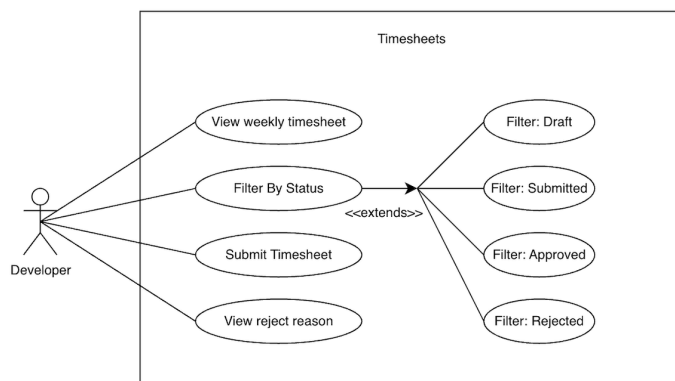
Discard timer

- TUCBW the user discards an active timer instead of stopping it.
- TUCEW the timer session is deleted and no time entry is created.

Edit / delete draft entry

- TUCBW the user selects a draft entry and chooses to edit or delete it.
- TUCEW the entry's changes are saved, or the entry is removed, provided its parent timesheet is still in Draft status.

Timesheets (Developer)



View weekly timesheet

- TUCBW the user navigates to the Timesheets screen.
- TUCEW the system displays the current week's entries grouped by task, with daily totals, a grand total and the timesheet's approval status.

Filter by status

- TUCBW the user selects a status filter (Draft, Submitted, Approved, or Rejected).
- TUCEW the system re-queries and displays only timesheets matching that status.

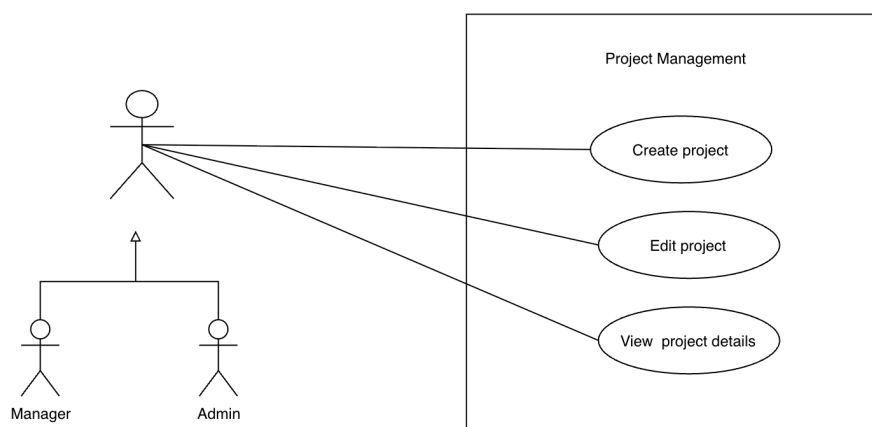
Submit timesheet

- TUCBW the user, viewing a Draft and unlocked timesheet, clicks "Submit Timesheet" and confirms.
- TUCEW the timesheet's status changes to Submitted, the week is locked from further editing and a success confirmation is shown.

View rejection reason

- TUCBW the user opens a timesheet in Rejected status.
- TUCEW the system displays the manager's written rejection reason inline alongside the entries.

Project Management



Create project

- TUCBW an admin or Manager enters the required project details and creates a new project.
- TUCEW the project is created withing the workspace and is available to for all members assigned to it

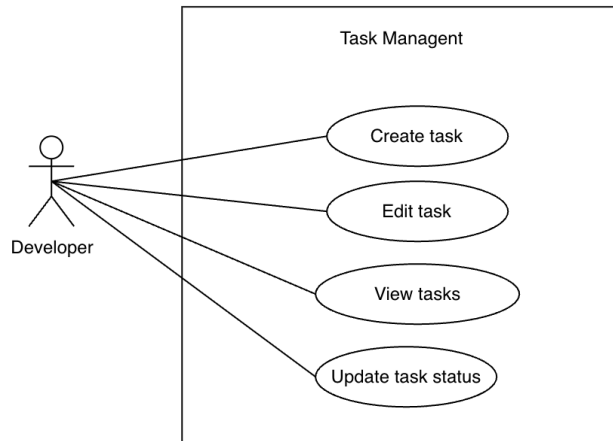
Edit project

- TUCBW an Admin or Manager opens an existing project and edits it's project details.
- TUCEW the updated details are saved and displayed.

View project details

- TUCBW an Admin or Manager opens a project to view its details
- TUCEW the system displays the project's information to the user.

Task Management (Developer)



Create Task

- TUCBW a developer or a Manager enters the required details to create a new task for a project.
- TUCEW the task is created and added to the selected project.

Edit task

- TUCBW a developer or manager open an existing task and edit its details.
- TUCEW the updated task details are saved and displayed.

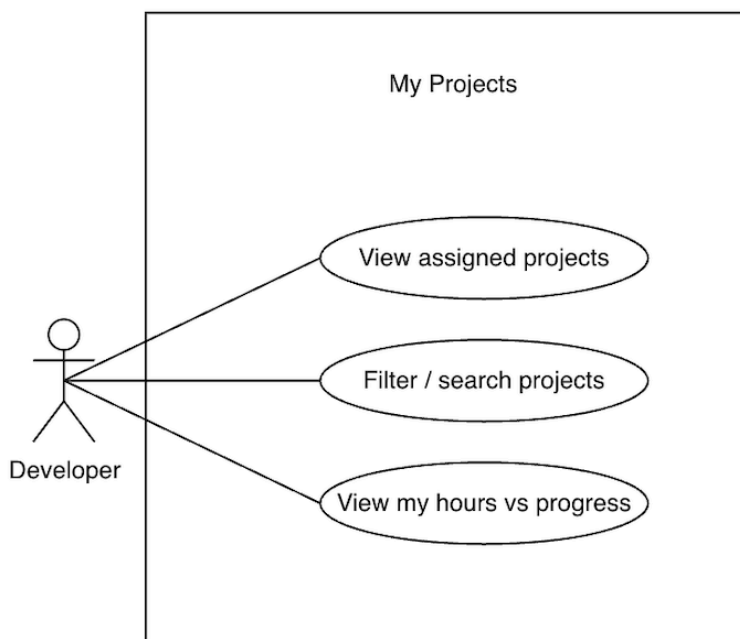
View my tasks

- TUCBW the user navigates to the My Tasks screen.
- TUCEW the system displays all tasks assigned to the user, with live counts by status.

Update task status

- TUCBW the user changes a task's status from the status dropdown.
- TUCEW the task's status updates on screen to reflect the new value.

Project Visibility (Developer)



View assigned projects

- TUCBW the user navigates to the Projects screen.
- TUCEW the system displays every project the user is personally a member of, as cards showing name and status.

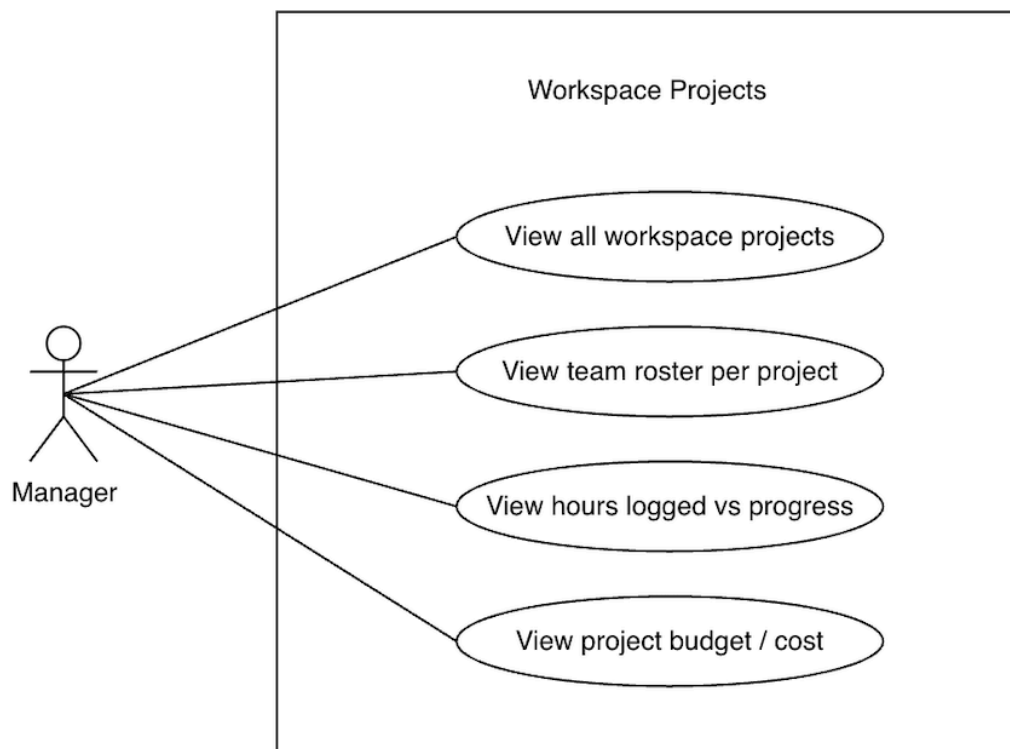
Filter / search projects

- TUCBW the user applies a status filter or types a project name into search.
- TUCEW the displayed project cards update to match the filter or search term.

View my hours vs. progress

- TUCBW a project card's detail call resolves after the initial list loads.
- TUCEW the card displays the user's own hours logged on that project, the project's overall progress percentage and the initials of its team members.

Project & Team Oversight (Manager)



View all workspace projects

- TUCBW a user with the Manager role navigates to the Projects screen.
- TUCEW the system displays every non-deleted project in the manager's workspace, regardless of whether the manager is personally assigned to it.

View team roster per project

- TUCBW a project card's detail call resolves for a manager.
- TUCEW the system displays every team member on that project alongside their individually logged hours.

View hours logged vs. progress

- TUCBW a project's detailed data.
- TUCEW the system displays total hours logged for the project and a progress bar (clamped to 100%) representing hours logged against budgeted hours.

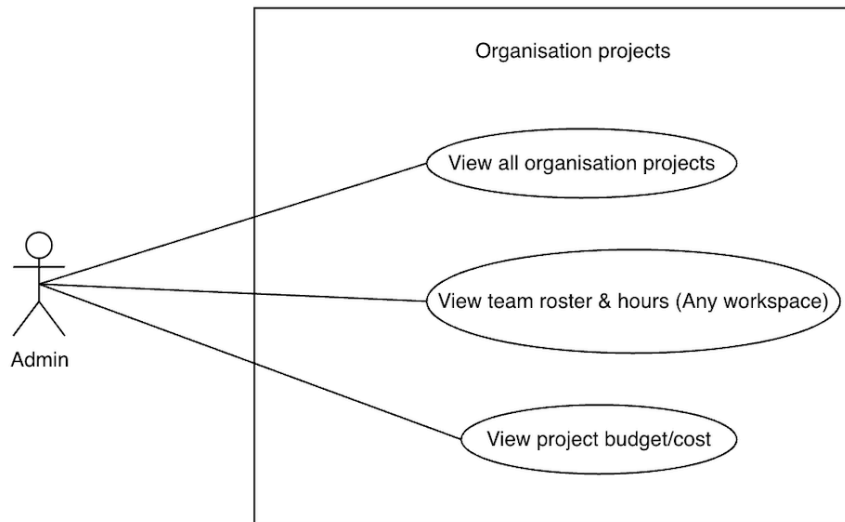
View project budget / cost

- TUCBW a manager views a project card or its details.
- TUCEW the system displays that project's budgeted hours, hourly rate and total cost, which are hidden from a Developer viewing the same project.

Manage team membership

- TUCBW a manager adds or removes a workspace member on the Team screen.
- TUCEW the member's workspace access is granted or soft removed and the change is reflected in project and task assignment availability.

Organisation Oversight (Admin)



View all organisation projects

- TUCBW a user with the Admin role navigates to the Projects screen.
- TUCEW the system displays every non-deleted project across every workspace in the organisation.

View team roster & hours (any workspace)

- TUCBW an admin views the details of any project, regardless of workspace.
- TUCEW the system displays that project's team roster and each member's hours logged, identically to the manager view.

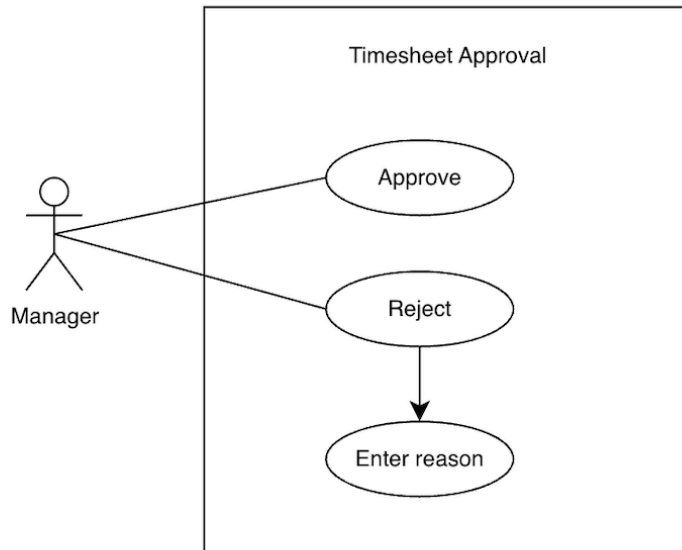
View project budget / cost

- TUCBW an admin views any project card or its details.
- TUCEW the system displays that project's budgeted hours, hourly rate and total cost.

View and act on organisation-wide timesheets and leave requests

- TUCBW an admin opens the timesheets or leave requests view.
- TUCEW every workspace's records are shown and the admin can approve or reject any of them, subject to the same self-approval restriction as a manager.

Timesheet Approval (Manager)



Approve

- TUCBW a manager, viewing a Submitted timesheet, clicks "Approve" and confirms.
- TUCEW the timesheet's status changes to Approved and is locked and a success confirmation is shown.

Reject

- TUCBW a manager, viewing a Submitted timesheet, clicks "Reject."
- TUCEW, after the manager enters a reason and confirms, the timesheet's status changes to Rejected with that reason stored and visible to the developer.

Enter reason

- TUCBW the manager opens the reject dialog.
- TUCEW a non-empty reason is captured and attached to the rejection before the request is sent.

Review with the AI manager assistant

- TUCBW a manager or admin opens the evidence review for a Submitted timesheet.

- TUCEW the system shows an evidence score and plain-English reasoning built from GitHub, Jira and calendar activity, highlighting missing or conflicting evidence. The assistant never decides, the manager still approves or rejects.

AI Insights & GitHub Integration (Developer/Manager)



View personal insights (Developer)

- TUCBW the user, in the Developer role, navigates to the Insights screen.
- TUCEW the system displays the user's weekly summary, productivity score, task-switching rate, delivery forecast and any flagged anomalies for their own logged time.

View team insights (Manager)

- TUCBW the user, in the Manager or Admin role, navigates to the Insights screen.

- TUCEW the system displays workspace-wide burnout risk flags, delivery forecasts per project and team productivity trends across all members.

Switch insights scope

- TUCBW the user selects "Overall" or "By project" from the scope switcher on the Insights screen.
- TUCEW the displayed insight cards and charts update to reflect only the selected scope.

Resolve an insight

- TUCBW a user marks an insight as resolved.
- TUCEW the insight is flagged resolved, with the resolver and time recorded and no longer appears as an open flag.

Connect GitHub account

- TUCBW the user selects "Connect GitHub" from the Insights screen, having no active GitHub connection.
- TUCEW the user is redirected to GitHub's OAuth consent screen and upon authorisation, the system stores the connection and marks the user as GitHub-connected.

Sync GitHub activity

- TUCBW a GitHub-connected user selects "Sync" on the Insights screen.
- TUCEW the system fetches the user's recent commit activity and refreshes the correlation between logged hours and GitHub contributions.

View hours-vs-commits comparison

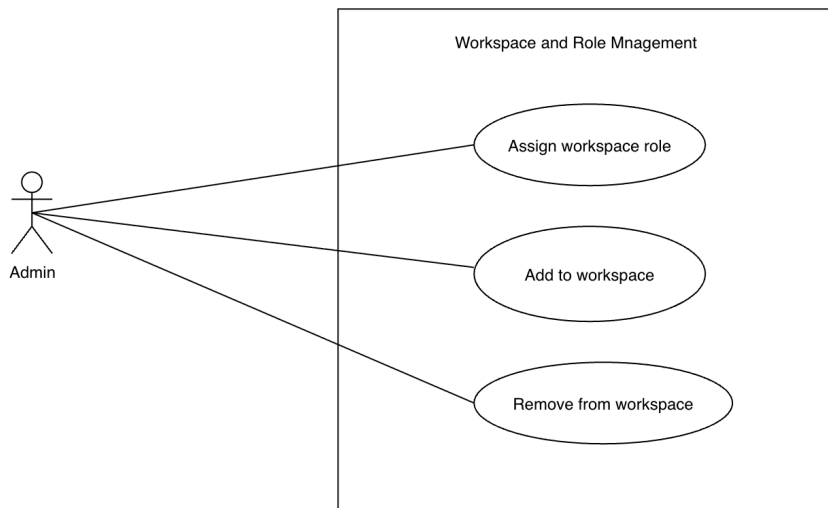
- TUCBW a GitHub-connected user views the "Git commits vs effort" insight.
- TUCEW the system displays a chart comparing logged hours against commit activity for the selected period, along with a correlation note.

Connect Jira / Google Calendar

- TUCBW the user selects "Connect" for Jira or Google Calendar, having no active connection for that provider.

- TUCEW the user completes the provider's OAuth consent flow and the connection is stored, unlocking that provider's tickets or events view and its contribution to the AI manager assistant's evidence.

Workspace & Role Management



Add user to workspace

- TUCBW anAdmin selects a user from the waiting room and add them to a workspace .
- TUCEW the user is added to a workspace

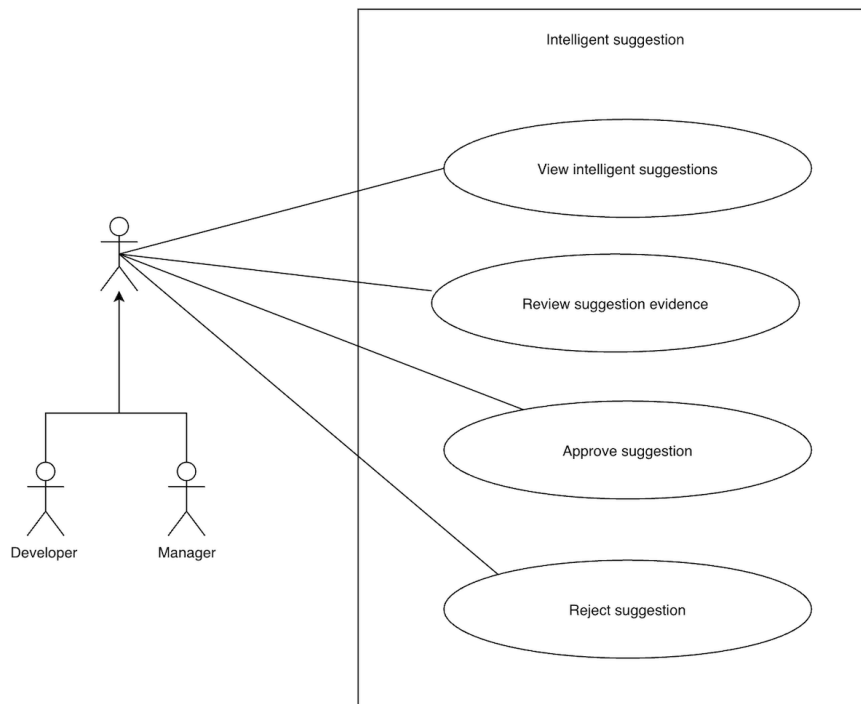
Assign workspace role

- TUCBW an admin adds a user to a workspace and selects a Developer, Manager or Admin role.
- TUCEW the user is assigned the selected role within the workspace.

Remove user from workspace

- TUCBW an admin selects an existing workspace member and chooses to remove them from the workspace.
- TUCEW the user is removed from the workspace and returns to the waiting room.

Intelligent Suggestion (wow factor)



View Intelligent Suggestions

- TUCBW a Developer opens Intelligent Suggestions.
- TUCEW the system displays AI-generated timesheet suggestions based on GitHub, Jira, calendar activity, and previous work patterns.

Review Suggestion Evidence

- TUCBW a Developer opens a suggested timesheet entry.
- TUCEW the system displays the supporting evidence and confidence for the suggestion.

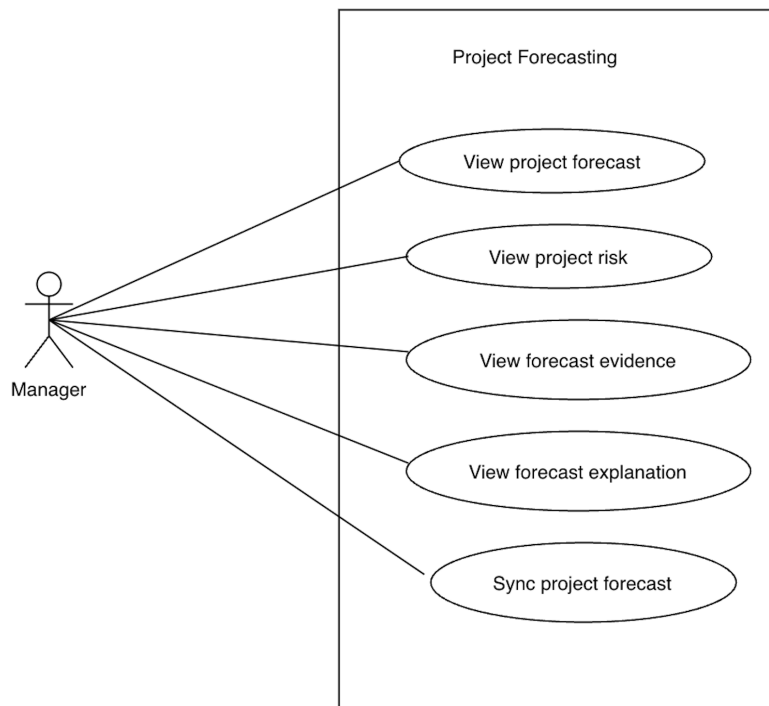
Approve or Edit Suggestion

- TUCBW a Developer reviews a suggestion and chooses to approve or edit it.
- TUCEW the confirmed suggestion is converted into a timesheet entry.

Reject Suggestion

- TUCBW a Developer reviews a suggestion and selects Reject.
- TUCEW the suggestion is rejected and is not added to the timesheet.

Project forecast (wow factor)



View project forecast

- TUCBW a Manager opens the Forecast section for the selected project
- TUCEW the system displays the latest project forecast, including used hours, estimated remaining work, projected total hours, expected completion date, project risk and forecast confidence.

View Project Risk

- TUCBW a Manager views the risk analysis for a project.
- TUCEW the system displays the project's budget, schedule and task-progress risks together with an overall status of Healthy, Warning, At Risk, or Unknown.

View Forecast Evidence and Confidence

- TUCBW a Manager views the evidence supporting a project's forecast.

- TUCEW the system displays the forecast confidence as High, Medium, or Low, together with the available and missing evidence used to determine that confidence.

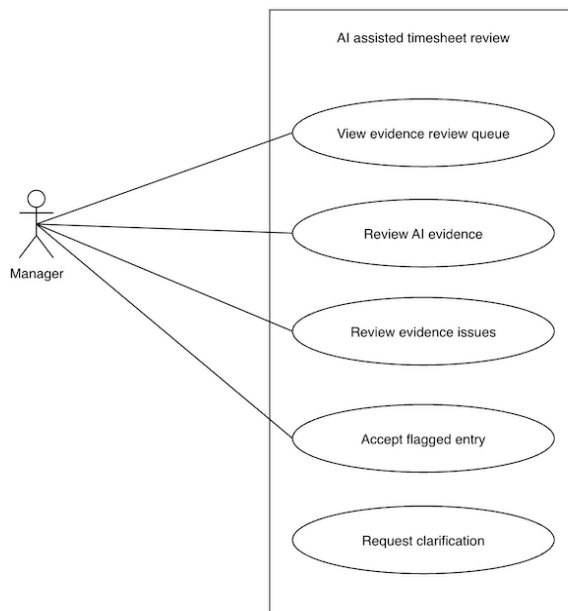
View AI Forecast Explanation

- TUCBW a Manager views the AI explanation associated with a project forecast.
- TUCEW the system displays an AI-generated summary explaining the calculated forecast, contributing risk factors and recommendations while acknowledging unavailable evidence.

Sync Project Forecast

- TUCBW a Manager selects Sync to update the forecast for a project.
- TUCEW the system recalculates the forecast using the latest project and supporting evidence, saves the updated forecast and displays the refreshed results and last-synced time.

AI-assisted Timesheet Review



View AI Evidence Review Queue

- TUCBW a Manager opens the Review Queue to review submitted timesheets.
- TUCEW the system prioritises flagged timesheets and highlights entries that require additional attention based on the available evidence.

Review AI Evidence

- TUCBW a Manager opens a flagged timesheet entry for review.
- TUCEW the system displays an AI evidence score together with relevant evidence from GitHub activity, Jira activity, calendar events, historical behaviour, and duration consistency.

Review Evidence Issues

- TUCBW a Manager reviews a timesheet entry where the AI Evidence Engine has identified unusual, missing, or conflicting evidence.
- TUCEW the system highlights the evidence that caused the entry to be flagged and explains why the entry may require further review without automatically determining that the entry is incorrect.

Accept Flagged Entry

- TUCBW a Manager reviews the timesheet entry and its supporting AI evidence and selects Accept.
- TUCEW the system accepts the entry and records the Manager's decision.

Request Clarification

- TUCBW a Manager reviews a flagged timesheet entry and selects Request Clarification when additional information is required.
- TUCEW the system returns the entry to the Developer for clarification while retaining the evidence associated with the flag.

Functional Requirements

FR1. Authentication & Session Management

FR1.1. Account access

- FR1.1.1. The system will allow a user to register a new account using a company email and password.
- FR1.1.2. The system will assign the Developer role to a newly registered account by default.
- FR1.1.3. The system will allow a user to log in with their company email and password and receive a signed session token (JWT).
- FR1.1.4. The system will attach the session token automatically to every subsequent authenticated request.
- FR1.1.5. The system will redirect an unauthenticated user attempting to access a guarded route to the login screen.
- FR1.1.6. The system will allow a user to log out, ending their active session

FR2. Time Logging

FR2.1. Manual time entry

- FR2.1.1. The system will allow a developer to select a project from only the projects they are assigned to.
- FR2.1.2. The system will allow a developer to select a task belonging only to the currently selected project.
- FR2.1.3. The system will calculate entry duration from the entered start and end time; duration will never be entered manually.
- FR2.1.4. The system will save a new manual entry with a status of Draft.
- FR2.1.5. The system will allow a developer to edit or delete a draft entry, provided its timesheet is not yet submitted or locked.

FR2.2. Timer-based time entry

- FR2.2.1. The system will allow a developer to start a timer against a selected project and task.
- FR2.2.2. The system will allow only one active timer per developer at any time.
- FR2.2.3. The system will allow a developer to pause and resume an active timer without loss of elapsed time.
- FR2.2.4. The system will automatically create a Draft time entry when a timer is stopped, requiring no further action from the developer.

FR3. Timesheets

FR3.1. Timesheet review

- FR3.1.1. The system will display a developer's current week's time entries grouped by task, with daily and weekly totals.
- FR3.1.2. The system will allow a developer to filter their timesheets by status (Draft, Submitted, Approved, Rejected).
- FR3.1.3. The system will display a manager's rejection reason inline on a rejected timesheet.

FR3.2. Timesheet submission

- FR3.2.1. The system will allow a developer to submit a Draft, unlocked timesheet for approval.
- FR3.2.2. The system will lock a timesheet from further editing immediately upon submission.
- FR3.2.3. The system will allow a rejected timesheet to be corrected and resubmitted.

FR3.3. Timesheet approval (Manager)

- FR3.3.1. The system will allow a user with the Manager role to approve a Submitted timesheet.
- FR3.3.2. The system will allow a user with the Manager role to reject a Submitted timesheet, requiring a non-empty written reason.
- FR3.3.3. The system will lock a timesheet once it has been approved or rejected.
- FR3.3.4. The system will restrict the Approve/Reject controls to users holding the Manager role only.

FR4. Task Management

FR4.1. Personal task list

- FR4.1.1. The system will display all tasks assigned to the current user.
- FR4.1.2. The system will allow the user to filter tasks by status (To Do, In Progress, Done, Blocked).
- FR4.1.3. The system will allow the user to search tasks by title.
- FR4.1.4. The system will display live counts of active, completed and blocked tasks.
- FR4.1.5. The system will allow the user to change a task's status.

FR5. Project Visibility & Oversight

FR5.1. Role-scoped project listing

- FR5.1.1. The system will display, to a Developer, only the projects that developer is personally assigned to.

- FR5.1.2. The system will display, to a Manager, every non-deleted project within that manager's workspace, regardless of personal assignment.
- FR5.1.3. The system will display, to an Admin, every non-deleted project across every workspace in the organisation.
- FR5.1.4. The system will allow the user to filter the displayed project list by status and search by project name.

FR5.2. Project detail

- FR5.2.1. The system will display, for a selected project, its full team roster and each member's individually logged hours.
- FR5.2.2. The system will display total hours logged for a project and a progress percentage against its budgeted hours, clamped between 0% and 100% for display.
- FR5.2.3. The system will display budgeted hours, hourly rate and total cost for a project only to users holding the Manager or Admin role.
- FR5.2.4. The system will deny project detail access to a user with no membership on that project and no Manager/Admin role.

Non-functional Requirements

All targets are quantified using an observability platform of choice: Spring Boot Actuator (Micrometer) for application metrics, Prometheus and Grafana for collection and dashboards, UptimeRobot for uptime monitoring and k6 for load generation.

1. Performance

The system should respond to user actions within 1-2 seconds under normal load, with appropriate UI feedback (loading indicators, disabled controls while a request is in flight).

Quantification:

- Normal load is 25 concurrent users.
- 95% of API requests complete within 2 seconds, with an error rate below 1%.

Measurement:

k6 for load testing, Spring Boot Actuator and Prometheus for application metrics, Grafana for dashboards.

Example test:

25 virtual users load projects, tasks and timesheets for 3 minutes. The test passes if p(95) is 2 seconds or less and fewer than 1% of requests fail.

2. Scalability

The system must scale to support at least 100 concurrent users without performance degradation.

Quantification:

At 100 concurrent users, 95% of API requests complete within 2 seconds, with an error rate below 1%.

Measurement:

k6 for load testing, with Prometheus and Grafana tracking CPU, memory and database connection pool usage.

Example test:

100 virtual users run the same workload for 3 minutes. The test passes if p(95) is 2 seconds or less and fewer than 1% of requests fail.

3. Availability

Core services must be available 99.9% of the time, excluding scheduled maintenance.

Quantification:

At least 99.9% availability, measured monthly. Container healthchecks and automatic restart policies support this target.

Measurement:

UptimeRobot monitoring the application health endpoint, with Spring Boot Actuator health checks and Prometheus uptime metrics.

Example test

A monitor checks the health endpoint every 5 minutes. Availability is reported as the percentage of successful checks over the monitoring period.

4. Recoverability

In case of total system failure, the platform must be fully restorable within 2 hours using automated fallback mechanisms.

Quantification:

- Time from total failure to the health endpoint reporting UP is 2 hours or less.
- No committed time entries or approved timesheets are lost (restored from the latest backup).

Measurement:

The automated deployment workflow, scheduled database backups and a timed recovery drill.

Example test:

Destroy the running stack on a test environment, rebuild it through the automated deploy path, restore the latest database backup and time how long it takes until the health endpoint reports UP. The test passes at 2 hours or less with all committed rows present.

5. Accessibility

The system must be usable by people with disabilities, including keyboard-only and screen-reader users, because time logging and approvals are daily workflows for every employee.

Quantification:

- Lighthouse accessibility score of at least 90 on the Login, Dashboard, Timesheets and Insights screens.
- 0 critical axe-core violations on the same four screens.

Measurement:

Chrome Lighthouse and axe DevTools, with Angular Material and semantic HTML as the design tactics.

Example test:

Log in as a seeded user, open each of the four screens and run Lighthouse (Accessibility) and an axe full-page scan. The test passes if every score is 90 or above and axe reports no critical violations.

Security

The system handles sensitive employee data (timesheets, productivity scores, personal details) and must protect it from unauthorised access, in line with POPIA.

Quantification:

- 100% of API endpoints containing protected data require authentication, except explicitly public endpoints.
- 100% of authorisation decisions are enforced server-side using Spring Security.
- Administrative accounts must not be able to access administrative functionality without successfully completing MFA
- 0 critical or high-severity vulnerabilities found by an OWASP ZAP baseline scan.
- SonarQube Cloud security rating of A on [main](#) (no open security issues of Medium severity or higher).
- Security review rating of A (100% of security hotspots reviewed).
- 0 open critical or high Dependabot alerts before a production release.

Measurement:

SonarQube Cloud (static analysis and hotspot review) and GitHub Dependabot (dependency scanning), alongside Spring Security, JWT and ZAP.

Example test:

Call every protected endpoint with no token and confirm each returns 401 or 403, then run a ZAP baseline scan against the deployed application. The test passes if all calls are rejected and ZAP reports no High findings.

Open the SonarQube Cloud dashboard for [main](#) and confirm the security rating is A and the security review rating is A and confirm the Dependabot tab shows no open critical or high alerts. The test passes if both hold.

6. Maintainability

The system must be easy to extend and test. As new features are added (additional external integrations, cost management, business owner features), the codebase must remain understandable and modifiable without breaking existing functionality. The architecture must support adding new external SSO providers or switching between them with minimal effort.

Quantification:

- At least 80% automated test coverage for backend business logic
- At least 70% automated test coverage for frontend business logic
- Adding a new external integration must require changes primarily within the relevant integration adapter package
- At least 90% of existing automated tests must continue to pass after the addition of a new integration
- SonarQube Cloud maintainability rating of A.
- Duplication of 3% or less on new code.
- Coverage reported through SonarQube Cloud (JaCoCo and Icov) against the 80% backend and 70% frontend targets.
- The quality gate must pass on every pull request. This gives your line "new features must not introduce new critical or high-severity issues" an actual enforcement point.

Measurement:

- JUnit 5
- Mockito
- SpringBoot Test
- Jest
- JaCoCo for Java code coverage

Example test:

Architecture should enforce that business logic must not directly depend on:

- Jira SDKs
- Google Calender API implementation classes
- GitHub API implementation classes
- SSO provider specific implementation classes

Instead adding a new integration just involves creating a new adapter instead of rewriting core business logic.

Look into using ArchUnit to fail a build if architectural boundaries are violated.

NFR Testing

ID	Requirement	Test / tool	Target	Actual so far
P-1	Normal load (25 users) responds in 1-2s	k6, 25 users, 3 min	p(95) ≤ 2s, errors < 1%	Met p(95) 1.34s, median 55ms, 0% errors
S-1	100 concurrent users without degradation	k6, 100 users, 3 min	p(95) ≤ 2s, errors < 1%	Met 0% errors across all runs. At 100 VUs, p(95) was 263.64ms for <code>/api/projects</code> and 281.62ms for the unread-count, active-timer and insights endpoints, essentially unchanged from 20 VUs, so no degradation under load.

A-1	99.9% monthly availability	UptimeRobot on the health endpoint	≥ 99.9%	Met , 99.188% Monitoring is ongoing; a full month is not yet observable, so this is an early reading, not a monthly figure.
R-1	Full restore within 2h	Recovery drill: destroy, rebuild via automated path, restore backup, time to health UP	≤ 2h, 0 rows lost	
ACC-1	Lighthouse accessibility ≥ 90 on 4 screens	Chrome Lighthouse	≥ 90	Met 95.3 SCORE for accessibility across 5 pages.
ACC-2	0 critical axe violations on 4 screens	axe DevTools	0 critical	Met 0 critical found
ACC-3	Static accessibility checks	SonarQube Cloud (accessibility tag)	0 open	Met 0 Open
SEC-1	Protected endpoints require authentication	Call each protected endpoint with no token	401/403 on all	Met
SEC-2	0 high/critical vulnerabilities	OWASP ZAP baseline scan	0 High	

SEC-3	Static analysis shows no medium+ security issues	SonarQube Cloud, main	Security rating A	Rating B
SEC-4	Hotspots reviewed	SonarQube Cloud	Security review rating A (100%)	Rating E, However upon further inspection, sonarqube does not take into consideration the existence of nginx
SEC-5	No high/critical dependency vulnerabilities	GitHub Dependabot	0 open high/critical	0 open
M-1	Quality gate passes on every PR	SonarQube Cloud quality gate	Gate passed	Met
M-2	Duplication and maintainability	SonarQube Cloud	Duplication ≤ 3%, maintainability A	Met Duplication 0.7% overall
M-3	Test coverage	JaCoCo and Jest, reported to SonarQube Cloud	≥ 70% backend, ≥ 70% frontend	

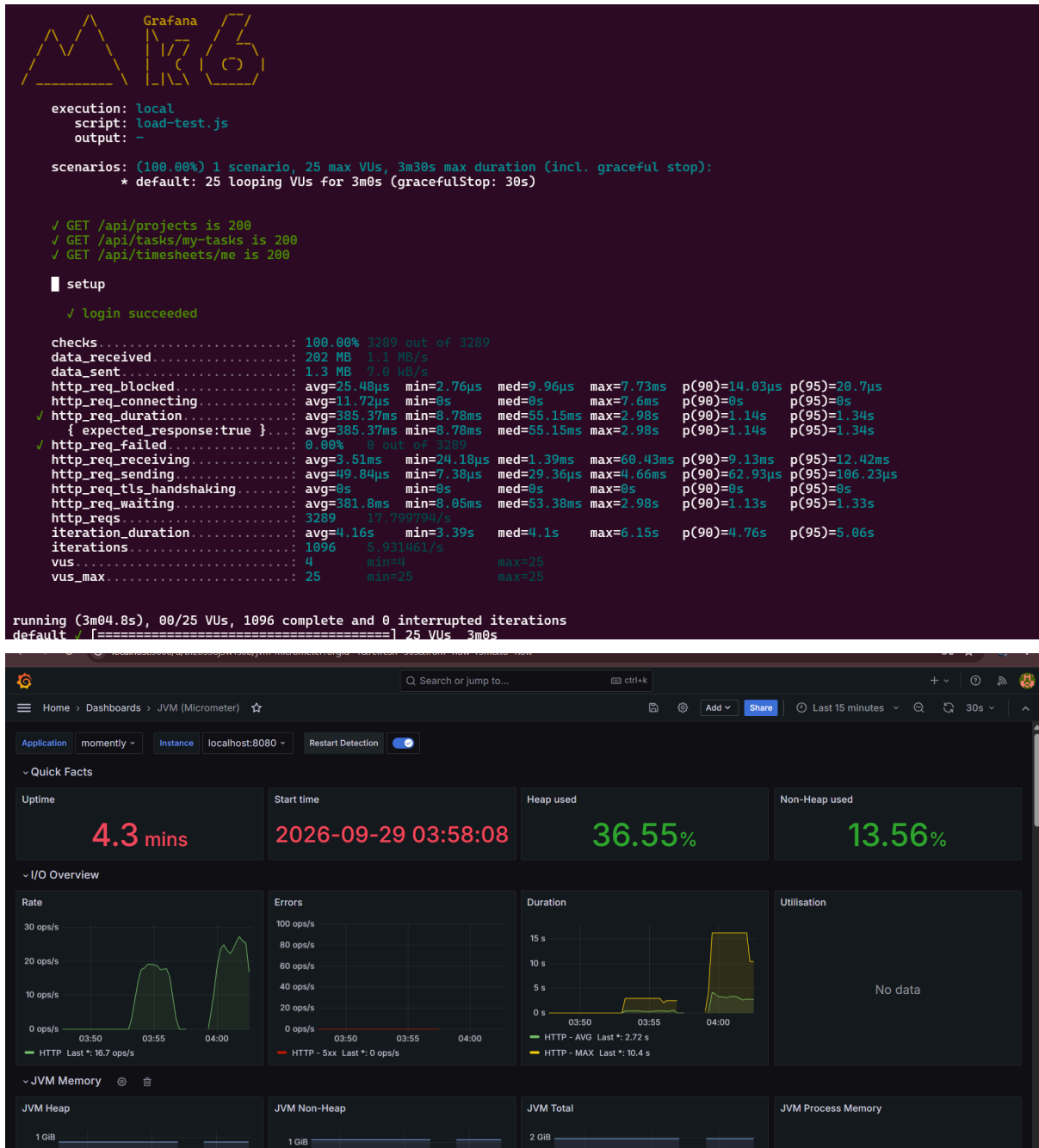
Bottleneck analysis (S-1)

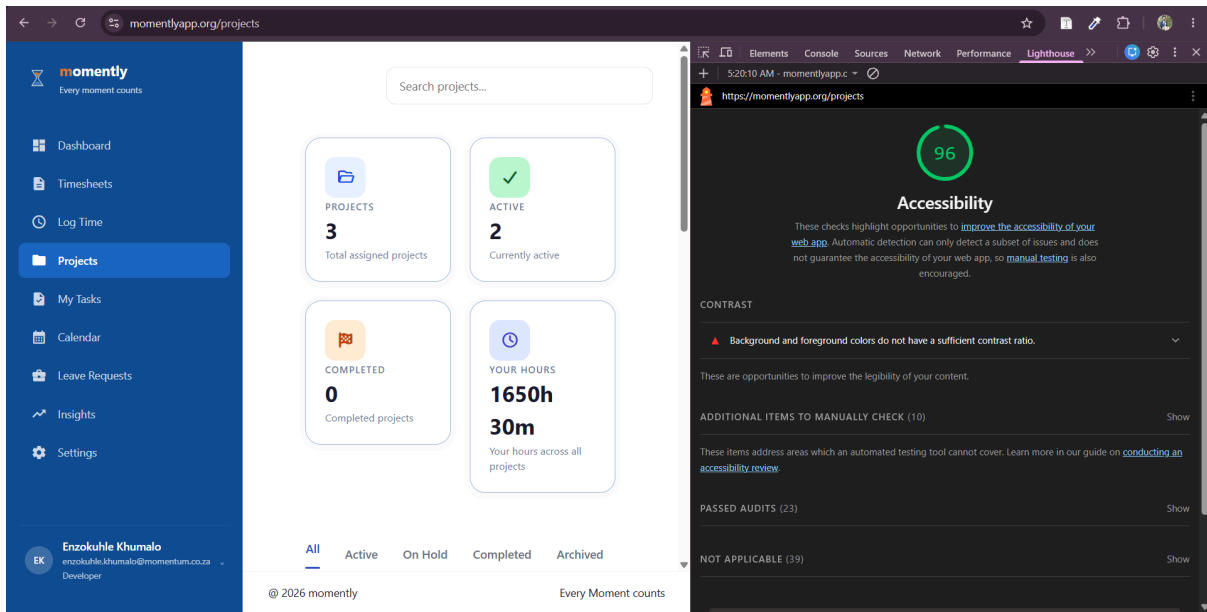
The first bottleneck at 100 users was the database connection pool. With the default HikariCP size of 10, requests queued waiting for a free connection. Raising it to 30 with `spring.datasource.hikari.maximum-pool-size` cut p(95) from 11.54s to 6.98s and removed the errors, but the 2 second target is still missed. The remaining suspected bottlenecks are unpaginated list endpoints (about 61 KB per response) and

no caching of read-heavy data. Each fix is applied one at a time with a k6 rerun so the effect of each change is visible.

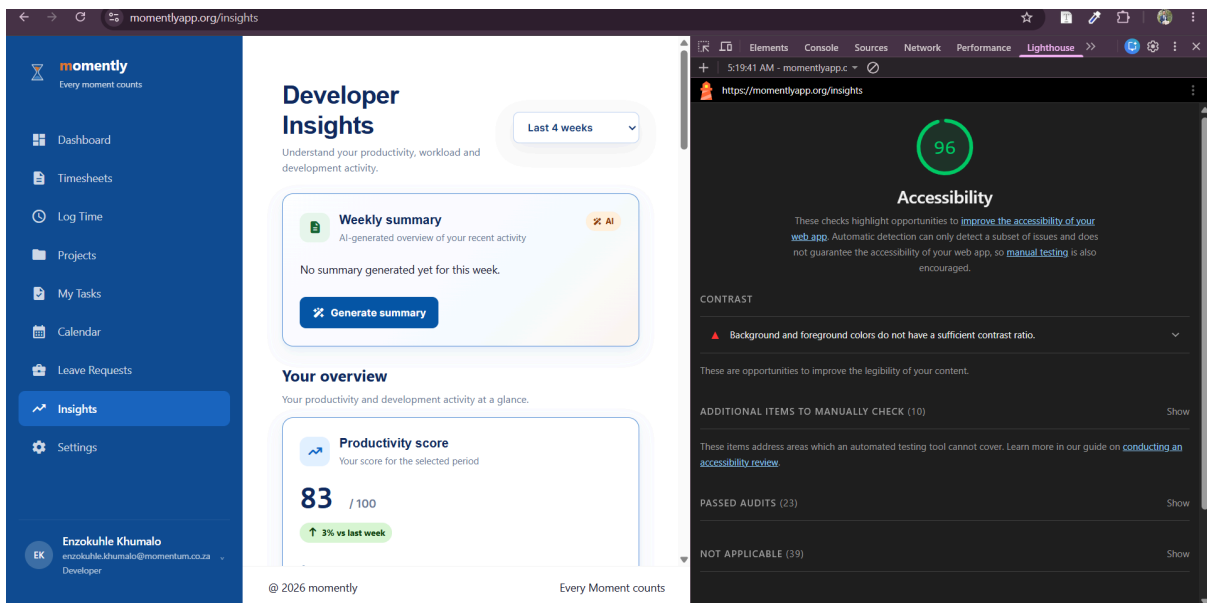
Root cause was an N+1 query pattern in `getMyTasks`, with a project lookup, access check and assignee lookup per task. **Fix: `batched lookups`**, with project names loaded in one `findAllById` query, access checked once per distinct project and the assignee name resolved once

Evidence





The screenshot shows the Momently Projects page. The left sidebar contains navigation links: Dashboard, Timesheets, Log Time, Projects (selected), My Tasks, Calendar, Leave Requests, Insights, and Settings. The main content area has a search bar and four project status cards: Projects (3), Active (2), Completed (0), and Your Hours (1650h 30m). The bottom of the main area shows filters: All, Active, On Hold, Completed, and Archived. The footer includes the user profile for Enzokuhle Khumalo and the text '@ 2026 momently Every Moment counts'. On the right, a Lighthouse accessibility audit is shown with a score of 96. The audit details include a contrast issue: 'Background and foreground colors do not have a sufficient contrast ratio.' and a list of additional items to manually check (10).



The screenshot shows the Momently Insights page. The left sidebar is the same as the Projects page. The main content area is titled 'Developer Insights' and includes a 'Weekly summary' section with a 'Generate summary' button. Below this is a 'Your overview' section showing a 'Productivity score' of 83 / 100, with a note that it is '3% vs last week'. The bottom of the main area shows the user profile for Enzokuhle Khumalo and the text '@ 2026 momently Every Moment counts'. On the right, a Lighthouse accessibility audit is shown with a score of 96. The audit details are the same as in the Projects page screenshot.

momently

Every moment counts

Dashboard

Timesheets

Log Time

Projects

My Tasks

Calendar

Leave Requests

Insights

Settings

EK

Enzokuhle Khumalo

enzokuhle.khumalo@momentum.co.za

Developer

Tue, Sep 29

Day

SUBMITTED

Submit Timesheet

The timesheet is awaiting review.

+ Add Manual Time Entry

Start Automatic Time Entry

Today's Entries

Total: 9h 30m

All

DRAFT

SUBMITTED

APPROVED

REJECTED

Project

All projects

From

09/29/2026

To

09/29/2026

Time	Project / Task	Type	Duration	Descrip
09:00 - 18:30	Mobile App Development	MANUAL	9h 30m	UI polis for dashb extend session

5:19:12 AM - momentlyapp.c

https://momentlyapp.org/log-time

91

Accessibility

These checks highlight opportunities to improve the accessibility of your web app. Automatic detection can only detect a subset of issues and does not guarantee the accessibility of your web app, so manual testing is also encouraged.

NAMES AND LABELS

Select elements do not have associated label elements.

These are opportunities to improve the semantics of the controls in your application. This may enhance the experience for users of assistive technology, like a screen reader.

CONTRAST

Background and foreground colors do not have a sufficient contrast ratio.

These are opportunities to improve the legibility of your content.

ADDITIONAL ITEMS TO MANUALLY CHECK (10)

Show

These items address areas which an automated testing tool cannot cover. Learn more in our guide on conducting an accessibility review

momently

Every moment counts

Dashboard

Timesheets

Log Time

Projects

My Tasks

Calendar

Leave Requests

Insights

Settings

EK

Enzokuhle Khumalo

enzokuhle.khumalo@momentum.co.za

Developer

Timesheets

View and manage your timesheets.

All

Draft

Submitted

Approved

Rejected

PERIOD	STATUS
Week 41 06 Oct 2025 - 12 Oct 2025	SUBMITTED
Week 42 13 Oct 2025 - 19 Oct 2025	SUBMITTED
Week 43 20 Oct 2025 - 26 Oct 2025	SUBMITTED
Week 44 27 Oct 2025 - 02 Nov 2025	SUBMITTED
Week 45 03 Nov 2025 - 09 Nov 2025	SUBMITTED
Week 46 10 Nov 2025 - 16 Nov 2025	SUBMITTED

5:18:48 AM - momentlyapp.c

https://momentlyapp.org/timesheets

100

Accessibility

These checks highlight opportunities to improve the accessibility of your web app. Automatic detection can only detect a subset of issues and does not guarantee the accessibility of your web app, so manual testing is also encouraged.

ADDITIONAL ITEMS TO MANUALLY CHECK (10)

Hide

Interactive controls are keyboard focusable

Interactive elements indicate their purpose and state

The page has a logical tab order

Visual order on the page follows DOM order

User focus is not accidentally trapped in a region

The user's focus is directed to new content added to the page

HTML5 landmark elements are used to improve navigation

Offscreen content is hidden from assistive technology

38

```

Grafana

execution: local
script: load-test.js
output: -

scenarios: (100.00%) 2 scenarios, 100 max VUs, 5m0s max duration (incl. graceful stop):
  * normal: 20 looping VUs for 2m0s (exec: run, gracefulStop: 30s)
  * peak: 100 looping VUs for 2m0s (exec: run, startTime: 2m30s, gracefulStop: 30s)

  ✓ /api/notifications/unread/count 2xx
  ✓ /api/timers/active 2xx
  ✓ /api/insights/me 2xx

  1 setup
  ✓ login succeeded

checks.....: 100.00% 11605 out of 11605
data_received.....: 6.4 MB 23 MB/s
data_sent.....: 508 kB 1.9 MB/s
http_req_blocked.....: avg=7.2ms min=35ns med=644ns max=1.8s p(90)=1.29µs p(95)=1.58µs
http_req_connecting.....: avg=2.14ms min=3s med=3s max=1.26s p(90)=3s p(95)=3s
http_req_duration.....: avg=247.25ms min=234.22ms med=239.51ms max=394.89ms p(90)=252.83ms p(95)=275.31ms
  { expected_response:true } : avg=247.25ms min=234.22ms med=239.51ms max=394.89ms p(90)=252.83ms p(95)=275.31ms
  ✓ { load:normal } : avg=241.26ms min=235.41ms med=238.94ms max=309.3ms p(90)=246.91ms p(95)=254.67ms
  ✓ { load:peak } : avg=248.44ms min=234.22ms med=239.65ms max=394.89ms p(90)=254.85ms p(95)=281.62ms
http_req_failed.....: 0.00% 0 out of 11605
http_req_receiving.....: avg=310.81µs min=13.54µs med=110.8µs max=142.55ms p(90)=189.49µs p(95)=283.8µs
http_req_sending.....: avg=148.97µs min=15.36µs med=107.7µs max=96.39ms p(90)=229.57µs p(95)=289.3µs
http_req_tls_handshaking.....: avg=5.05ms min=3s med=3s max=1.22s p(90)=3s p(95)=3s
http_req_waiting.....: avg=246.8ms min=234.01ms med=239.16ms max=323.28ms p(90)=252.18ms p(95)=274.35ms
http_reqs.....: 11605
iteration_duration.....: avg=3.76s min=3.7s med=3.72s max=0.52s p(90)=3.8s p(95)=4.01s
iterations.....: 3868 10.059217/s max=100 min=100
vus.....: 9 min=9 max=100
vus_max.....: 100 min=100 max=100

running (4m35.1s), 000/100 VUs, 3868 complete and 0 interrupted iterations
normal / [=====] 20 VUs 2m0s
peak / [=====] 100 VUs 2m0s

```

UptimeRobot

Monitoring

3rd party services

Incidents

Status pages

Maintenance

Team members

Integrations & API

Cybernauts...

Upgrade now

Monitoring

momentlyapp.org

HTTP/S monitor for <https://momentlyapp.org/>

Test Notification

Pause

Edit

Current status

Up

Currently up for 18h 20m 28s

Last check

1m, 19s ago

Checked every 5m [Get 60 sec. checks](#)

Last 24 hours

98.775%

2 incidents, 15m, 53s down

Domain & SSL

Domain valid until

Unlock

SSL certificate valid until

Unlock

Available only in Solo, Team and Scale [Upgrade now](#)

Next maintenance

No maintenance planned.

Set up maintenance

Regions

Last 7 days

98.775%

2 incidents, 15m, 53s down

Last 30 days

98.775%

2 incidents, 15m, 53s down

Last 365 days

--.---%

[Unlock with paid plans](#)

Pick a dat...

MTBF 7 days

10.68 hours

Response time for All regions

Setup alerts For slow response times

Last hour

97ms

50ms

25ms

sonarqubecloud Bot commented 2 hours ago

Quality Gate passed

Issues

0 New issues

0 Accepted issues

Measures

0 Security Hotspots

0.0% Coverage on New Code

0.0% Duplication on New Code

See analysis details on SonarQube Cloud

Maintainability snapshot

Maintainability rating

Overall code

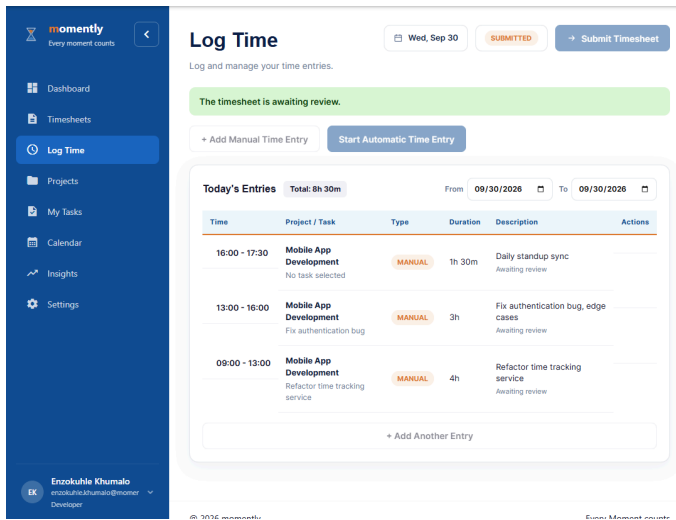
A

Reliability snapshot

Reliability rating

Overall code

A



Log Time
Log and manage your time entries.

The timesheet is awaiting review.

+ Add Manual Time Entry Start Automatic Time Entry

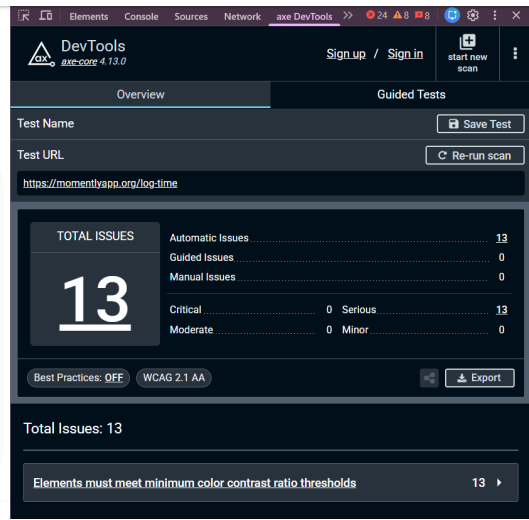
Today's Entries Total: 8h 30m From 09/30/2026 To 09/30/2026

Time	Project / Task	Type	Duration	Description	Actions
16:00 - 17:30	Mobile App Development No task selected	MANUAL	1h 30m	Daily standup sync Awaiting review	
13:00 - 16:00	Mobile App Development Fix authentication bug	MANUAL	3h	Fix authentication bug, edge cases Awaiting review	
09:00 - 13:00	Mobile App Development Refactor time tracking service	MANUAL	4h	Refactor time tracking service Awaiting review	

+ Add Another Entry

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axe DevTools
axe-core 4.13.0

Sign up / Sign in start new scan

Overview Guided Tests

Test Name Save Test

Test URL Re-run scan
https://momentlyapp.org/log-time

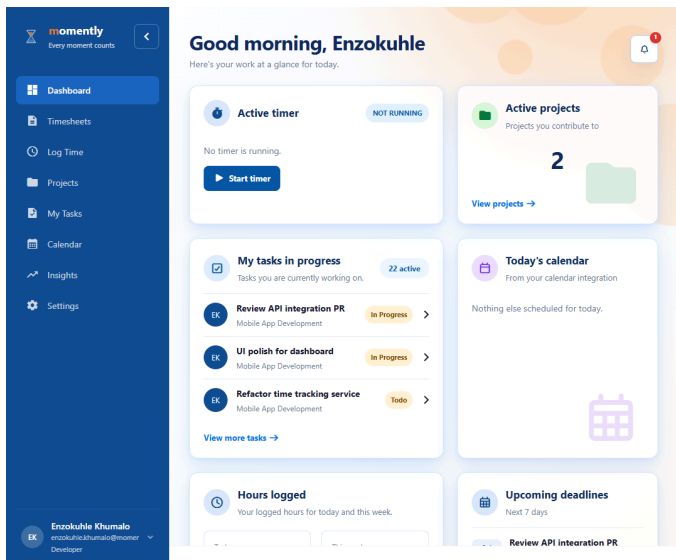
TOTAL ISSUES
13

Issue Type	Count
Automatic Issues	13
Guided Issues	0
Manual Issues	0
Critical	0
Serious	13
Moderate	0
Minor	0

Best Practices: OFF WCAG 2.1 AA Export

Total Issues: 13

Elements must meet minimum color contrast ratio thresholds 13



Good morning, Enzokuhle
Here's your work at a glance for today.

Active timer NOT RUNNING
No timer is running. Start timer

Active projects
Projects you contribute to
2 View projects →

My tasks in progress 22 active
Tasks you are currently working on.

- Review API integration PR Mobile App Development In Progress
- UI polish for dashboard Mobile App Development In Progress
- Refactor time tracking service Mobile App Development Todo

View more tasks →

Today's calendar
From your calendar integration
Nothing else scheduled for today.

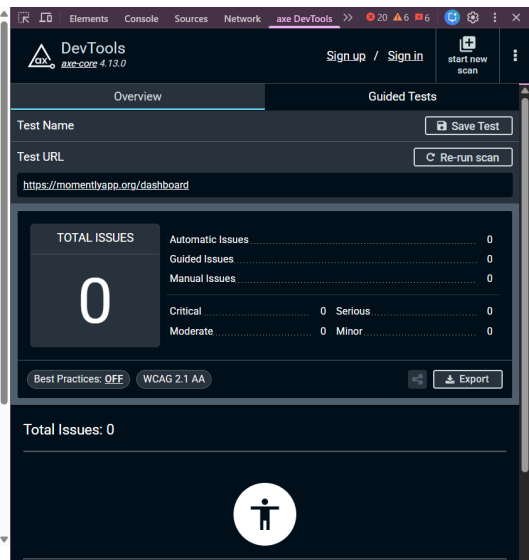
Hours logged
Your logged hours for today and this week.

Upcoming deadlines
Next 7 days

Review API integration PR

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Developer

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axe DevTools
axe-core 4.13.0

Sign up / Sign in start new scan

Overview Guided Tests

Test Name Save Test

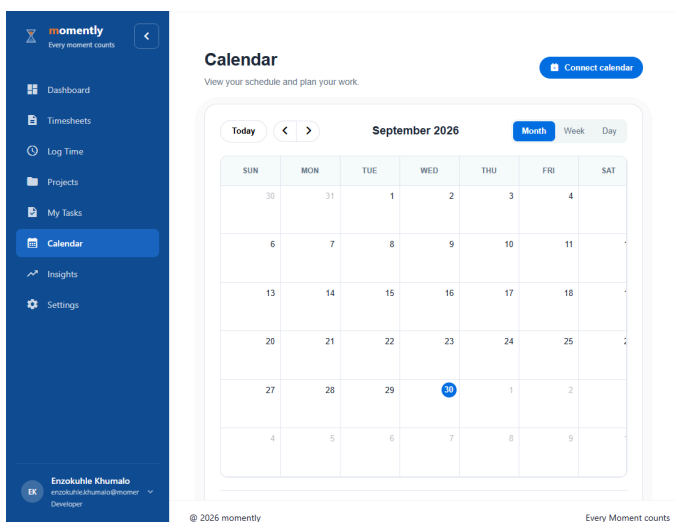
Test URL Re-run scan
https://momentlyapp.org/dashboard

TOTAL ISSUES
0

Issue Type	Count
Automatic Issues	0
Guided Issues	0
Manual Issues	0
Critical	0
Serious	0
Moderate	0
Minor	0

Best Practices: OFF WCAG 2.1 AA Export

Total Issues: 0



Calendar
View your schedule and plan your work.

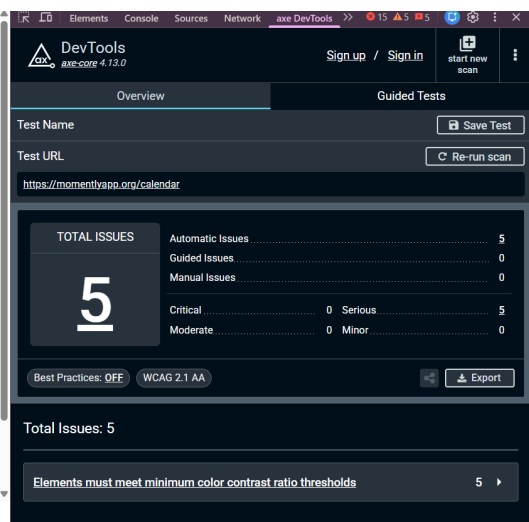
Connect calendar

Today September 2026 Month Week Day

SUN	MON	TUE	WED	THU	FRI	SAT
30	31	1	2	3	4	
6	7	8	9	10	11	
13	14	15	16	17	18	
20	21	22	23	24	25	
27	28	29	30	1	2	
4	5	6	7	8	9	

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axe DevTools
axe-core 4.13.0

Sign up / Sign in start new scan

Overview Guided Tests

Test Name Save Test

Test URL Re-run scan
https://momentlyapp.org/calendar

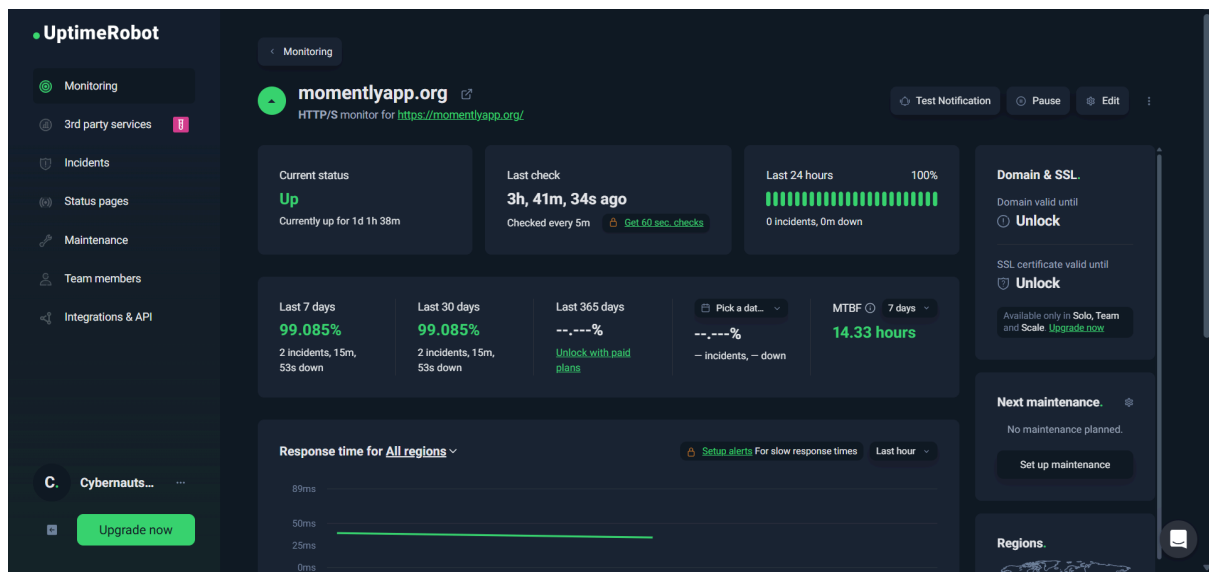
TOTAL ISSUES
5

Issue Type	Count
Automatic Issues	5
Guided Issues	0
Manual Issues	0
Critical	0
Serious	5
Moderate	0
Minor	0

Best Practices: OFF WCAG 2.1 AA Export

Total Issues: 5

Elements must meet minimum color contrast ratio thresholds 5



Domain Model

